

Charge Trapping and Conduction in γ Irradiated Isotactic Polypropylene

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ABSTRACT

The present work aims at investigating the relationship between the structural change and the electrical properties of isotactic polypropylene irradiated by gamma-rays at doses up to 100 kGy. Physicochemical analyses revealed a substantial increase of the carbonyl index of the material after irradiation along with a considerable increase of crystallinity which can be attributed to the cross-linking phenomenon. Dielectric properties of the materials were probed by charging/discharging currents and thermally stimulated currents. A significant increase in the conductivity was observed in comparison to the virgin sample. The thermally stimulated discharge current exhibits two peaks, whereas no peak was detected in virgin sample. These features appear related to the detrapping of space charges from deep traps created upon irradiation.

Index Terms — Polypropylene, space charge, gamma radiation, deep trap.

1 INTRODUCTION

POLYMERIC insulating materials integrated in electronic and electrical devices also find applications in space equipments and nuclear power stations. In this case, the materials are exposed to high energy radiation like gamma rays produced by radioactive elements. When such ionizing radiations interact with a polymer, it changes the structural arrangement. The alteration of the structure might occur by chemical radical reactions as a result of chain scission, oxidation and cross-linking mechanisms. The chain scission and oxidation process help to form smaller molecules, and hence facilitates dipoles orientation. On the other hand, the cross-linking reaction helps to form a three-dimensional structure within the material, preventing thus dipole orientation. The radiation not only alters the physical and chemical structure of the polymer. Indeed, it leads to the creation of defects in the polymer and can therefore enhance the presence of trapped charges and impact on the insulation properties [1]. Several experimental techniques have been implemented to understand the changes in electrical properties of polymeric materials (e.g. Poly(ethylene terephthalate) – PET, Poly(ethylene naphthalate) –PEN, low density Polyethylene –LDPE, etc.) after irradiation [2-5]. The isothermal charging (ICC), and discharging (IDC) currents along with thermally stimulated discharge current (TSDC) are among the most used methods to analyze charging properties

of insulating materials [6, 7]. It was reported that the measured current increases after exposure to gamma radiations [5, 8]. However, the relation between structural alterations and electrical properties of polymers are not yet fully understood. In this contribution, our purpose is to correlate the structural change with the electrical properties of the gamma irradiated Polypropylene -PP. PP has been chosen for its relatively simple chemical formulation along with for its ~~and~~ wide application to capacitors. On the other hand, owing to extensive application in packaging [9], the impact of radiation has been investigated from the standpoint of mechanical properties (embrittlement), and only scarce reports concern the impact on its dielectric properties [10]. In this work, the molecular modifications have been investigated by means of differential scanning calorimetry (DSC) and infrared spectroscopy (FTIR). Specific attention is paid to the impact of gamma irradiation on the transient charging current of PP. So-called final thermally stimulated discharging current – FTSDC measurements are also implemented to provide qualitative information concerning the density and depth of charge traps created after irradiation.

2 EXPERIMENTAL

2.1 SAMPLES PREPARATION

The samples used in this study are from a commercially available bi-axially oriented isotactic polypropylene (iPP) film of 25 μm in thickness characterized earlier [11, 12]. The samples were irradiated in air at room temperature using ^{60}Co

Manuscript received on 8 February 2017, in final form 24 July 2017, accepted 7 August 2017. Corresponding author: N. Saidi-Amroun.

gamma source with a dose rate of 42 Gy/min at the Nuclear Research Center of Algiers. The total dose of irradiation was 100 kGy. The samples were prepared by the thermal evaporation of gold electrodes (2 cm in diameter) on both sides of the samples surface in order to get good electrical contacts.

2.2 PHYSICAL AND CHEMICAL ANALYSES

FTIR spectra of the irradiated and pristine iPP samples were recorded in the transmission mode using a Thermo-Nicolet Nexus 670 spectrometer. All the measurements were performed in the $[4000 - 400 \text{ cm}^{-1}]$ range with a resolution of 2 cm^{-1} . The main effects of irradiation on the characteristic absorbance bands are discussed.

The thermal analyses of virgin and irradiated iPP were obtained using TA Instruments differential scanning calorimeter (DSC Q10). DSC thermograms were recorded from 2 to 175°C with a constant heating rate of 10°C/min on samples of c.a. 10 mg enclosed in aluminum pans. The cooling process was carried out from 175°C to room temperature with a constant cooling rate of 10°C/min under a nitrogen atmosphere to avoid the thermal degradation. The obtained data were analyzed by TA Universal Analysis software.

2.3 ELECTRICAL CHARACTERIZATION

For current measurements, the samples were placed between two metal electrodes in the measurement cell [13] which is connected to a vacuum pump and a gas cylinder. Measurements were realized in nitrogen at atmospheric pressure. The DC voltage was supplied by a Hewlett Packard stabilized power supply, and the current was measured by a Keithley 617 programmable electrometer. Figure 1 summarizes the measurement protocol as a function of time during the three steps. For the isothermal charging (ICC) and discharging (IDC) current measurements, the temperature was set to T_C value at which, the measurements have been carried out. Once the temperature was stabilized the ICC current was measured by applying a DC electric field E for a time t_c . Thereafter, the sample was short-circuited for the same time t_d and the current (IDC) was recorded. The measurements of the thermally stimulated discharging current, or also called final thermally stimulated discharging current (FTSDC) in what follows, were performed during the

sample heating process from T_C (the charging temperature) to T_F (the final temperature), with a constant heating rate b of 3°C/min . These steps were repeated for different temperature values T_C , T_F , field, and time t_c . The same sample was used for the different experimental conditions. T_F does not exceed 120°C , i.e. substantially below the melting point. In addition, the sample was kept short-circuit for at least 10 h at room temperature between consecutive protocols. In these conditions, we expect negligible effect of the previous thermo-electrical stresses on the material response.

3 RESULTS

3.1 FTIR ANALYSIS

The change in the chemical structure of polypropylene due to gamma irradiation effect was investigated by FTIR spectroscopy. The FTIR spectra of virgin and irradiated iPP films are shown in Figure 2, focusing on the region where the carbonyl groups appear ($750\text{--}1800 \text{ cm}^{-1}$). All the bands identified from the spectra are the bands characteristic of the virgin PP indicating the isotactic nature of the polymer. The band at 1453 cm^{-1} (high absorbance) is attributed to the asymmetric deformation in C-H of the CH_3 groups and symmetrical shear in C-H of the CH_2 groups. The peaks that appear at around 1375 cm^{-1} and 1356 cm^{-1} , are assigned respectively to the C-H and C-C bonds of the hydrocarbon chain. A band at 1169 cm^{-1} was attributed to the elongation vibrations of the C-C bond of the ethyl group C-CH_3 . The most important change after irradiation occurs at 1730 cm^{-1} indicating the formation of carbonyl groups, corresponding to the production of aldehydes and ketones. The calculated carbonyl index, which corresponds to the ratio between the area of the absorption band of C=O and the area of the band at 1169 cm^{-1} [14] are 1.5 and 4.5 % for the virgin and irradiated iPP respectively. This probably results from the interaction of oxygen molecules with the free radicals, which are trapped in the PP [15, 16]. This is all the more likely that the irradiation was achieved in air. The mechanism of formation of the alcohol or ketone groups is not very clear. Researches have revealed that, the amorphous-crystalline boundary domain plays an important role in the resistance to radiation-induced

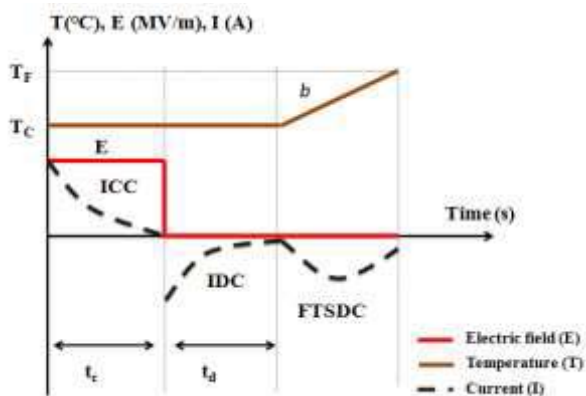


Figure 1. The measurement protocol showing the variation of temperature, electric field and current with time during the three steps. Typical values of t_c and t_d used in this work are 2000 to 3000 s.

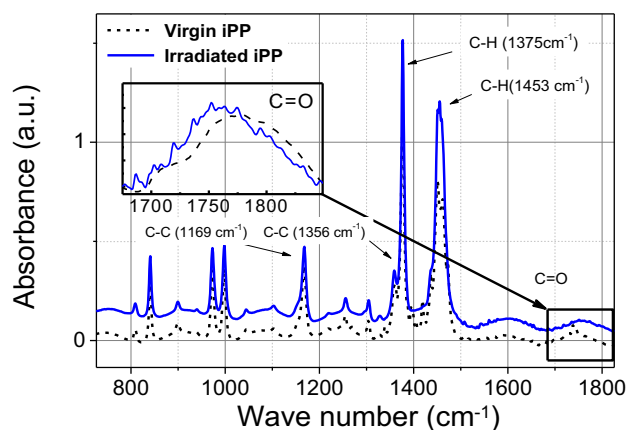


Figure 2. FTIR spectra of virgin and irradiated iPP in the wavenumber region $750\text{--}1800 \text{ cm}^{-1}$

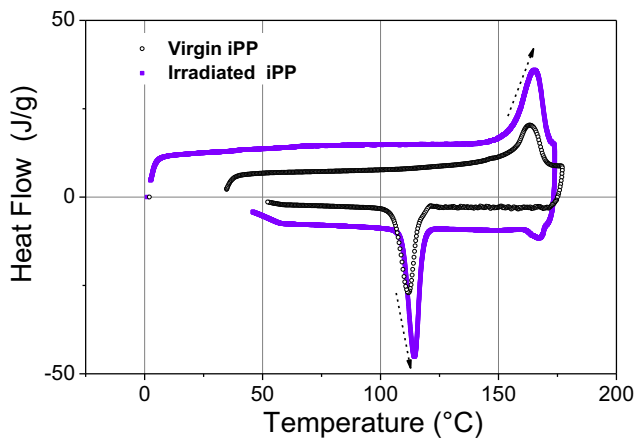


Figure 3. DSC thermograms of virgin and irradiated iPP obtained under heating/cooling rates of 10 °C/min.

Table 1. Parameters extracted from DSC thermograms

Parameters	Virgin iPP	Irradiated iPP
$T_m(^{\circ}\text{C})$	163	165
$T_c(^{\circ}\text{C})$	111	114
$\Delta H_m (\text{J/g})$	108	135
$\Delta H_c (\text{J/g})$	106	125
$\chi (\%)$	51.67	64.51

oxidation. In contrast, the oxidation occurs mainly in the amorphous area while the crystalline phase is scarcely oxidized [17, 18].

3.2 DSC ANALYSIS

Figure 3 depicts the DSC thermograms of the virgin and irradiated iPP. The degree of crystallinity (χ) was calculated using the following relation:

$$\chi(\%) = \frac{\Delta H_m}{\Delta H_0} \times 100 \quad (1)$$

where ΔH_m is the area under the DSC thermogram curve and ΔH_0 is the enthalpy of melting for a fully crystalline polymer and was estimated to be 209 J/g [19]. The melting temperature, crystallization temperature, melting enthalpy, enthalpy of crystallization, and the degree of crystallinity are summarized in Table 1.

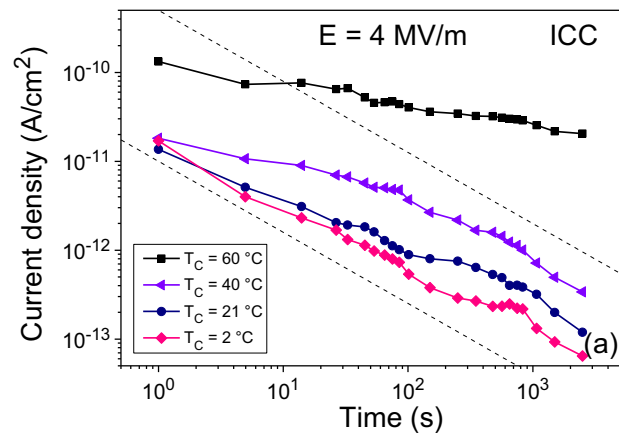
The melting enthalpy and melting temperature tend to increase upon irradiation. This is generally attributed to the growth of the crystalline lamellae [20, 21]. The increase of the melting enthalpy can be unambiguously attributed to the higher crystallinity degree in the polymer upon irradiation. This increase of the crystallinity compared to the virgin sample may be due to: (1) the cross-linking, which caused an increase in the branching degree or (2) the formation of new lamellae by re-crystallization of small fractions, which are produced by chains scission during irradiation.

In this case, the cross-linking occurs essentially in the amorphous region, because the rigid structure of the crystalline zones limits the probability of association in two

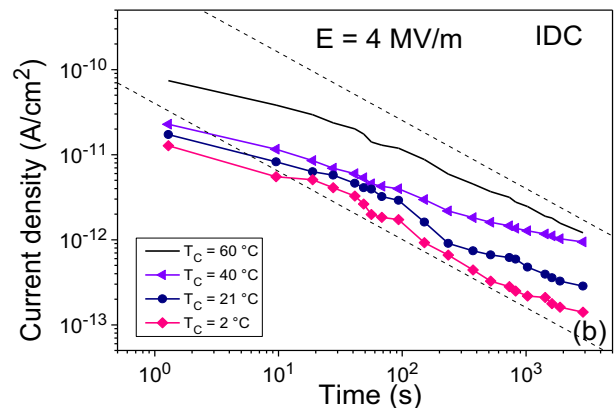
free radicals [22]. The crystallization begins with lamellae formation followed by the organization that creates the spherulites [23]. The increase of crystallinity has already been observed experimentally on a gamma-irradiated isotactic polypropylene by Stojanovic et al. [24] and Schawe [25]. In the case of irradiation in air, an oxidative degradation can be enhanced. On the other hand, Spadaro et al [26] have studied the molecular modifications induced in an isotactic PP irradiated in vacuum, where the total absorbed doses were in the range 6 - 1500 kGy. They have concluded that chain scission is the main effect at relatively low absorbed doses (< 300 kGy).

3.3 ISOTHERMAL CHARGING AND DISCHARGING CURRENTS

Figure 4 (a and b) illustrates the variation of the charging and the discharging currents versus time for a field $E = 4 \text{ MV/m}$ and for different charging temperatures T_C (2, 21, 40 and 60 °C) of the irradiated polypropylene. The obtained results for $t_c = t_d = 3000 \text{ s}$, present a monotonous decrease of the current according to the Curie -Von Schweidler law ($J = J_0 t^{-n}$) [27]. The slope of the charging current in case of virgin iPP was $n = 0.8$, and it was pretty the same for the discharge current [28]. The reverse power law with $n = 0.8$ is drawn in Figure 4. It shows clearly that the current decay, either ICC or IDC, is slower for irradiated than for neat iPP.



(a) Charging current density



(b) Discharging current density

Figure 4. The plot of isothermal charging and discharging current in iPP for different temperatures T_C . Charging and discharging times are 3000 s. The dashed lines are power laws with $n = 0.8$.

For the IDC, in average, the slope for irradiated iPP is 0.58 ± 0.10 . For ICC it increases from 0.22 to 0.58 as the temperature is decreased from 60 to 2 °C. The decrease of the slope with increasing temperature may come in part from a contribution from conduction processes to the charging current. For the IDC, there is no clear trend regarding the evolution of n with temperature. As it can be seen, the current magnitude increases substantially with increasing T_C . The steady state was not established within the 3000 s charging time. The same features were observed for virgin iPP [28], however, the current values in the irradiated iPP are up to four orders of magnitude higher. Figure 5 shows a comparison of the charging and discharging currents obtained at 100 °C under 6 MV/m for neat and irradiated iPP. The strong difference in current magnitude as well as in the shape of the transient current mentioned previously is confirmed.

The estimated conductivity at $T_C = 100$ °C is $\sigma = 3.6 \times 10^{-16} \Omega^{-1} \cdot m^{-1}$ for virgin iPP and $3.1 \times 10^{-12} \Omega^{-1} \cdot m^{-1}$ for the irradiated sample. Figure 6 depicts the conductivity as a function of the reciprocal temperature revealing an Arrhenius law for the dependence of conductivity vs. temperature:

$$\sigma = \sigma_0 \exp\left(\frac{-E_a}{kT_C}\right) \quad (2)$$

where k is the Boltzmann's constant and T_C is the absolute temperature. Remind that current data used here are that measured at the end of the charging step for 3000 s. As the steady state current is not reached after that time, as often with dielectrics, conductivity values should be considered as quasi-steady state. The apparent activation energy E_a of the current is estimated as ~ 0.71 eV for irradiated iPP, i.e. substantially higher than for virgin iPP (0.43 eV from [28]).

As shown above and consistently with earlier reports [28, 29], the irradiation in air enhances the oxidation rate and consequently increases the carbonyl density. Presumably, the electrical conductivity increases in the irradiated iPP due to the oxidation phenomenon.

An anomalous behavior of the charging current was observed at relatively high charging temperatures (60, 80 and 100 °C) and for different electric field strengths (2, 4 and 6 MV/m). We present in Figure 7a, the variation of the current vs. time in linear scale (not visible in logarithmic scale) at $T_C = 100$ °C and $E = 6$ MV/m. As it can be seen, current peaks appear in regular time intervals. Such phenomenon was not observed in the virgin iPP or in other polymers characterized with identical devices and in the same conditions as was reported in our previous works [13, 28]. Similar anomalous current was obtained by Many and Rakavy and was attributed to the SCLC (Space Charge Limited Current) at the particular value of the transit time of injected charges to cross the insulation thickness [30]. Several workers have observed these peaks in other polymers such as polyethylene (PE) [31] and polyethylene terephthalate (PET) [32]. The current peaks appear roughly periodically in time. Therefore, in order to analyze empirically this current behavior (Figure 7a) with extraction of the characteristic time of the process, the current was decomposed into monotonous and anomalous components (Figure 7b), using the following sinusoidal fit function:

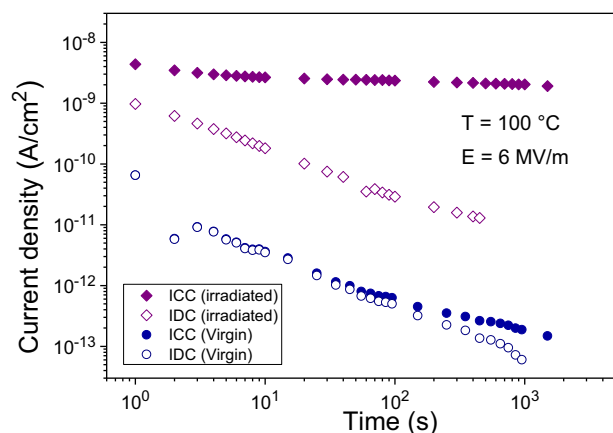


Figure 5. Charging and discharging current transients at 100 °C and 6 MV/m for neat and irradiated iPP.

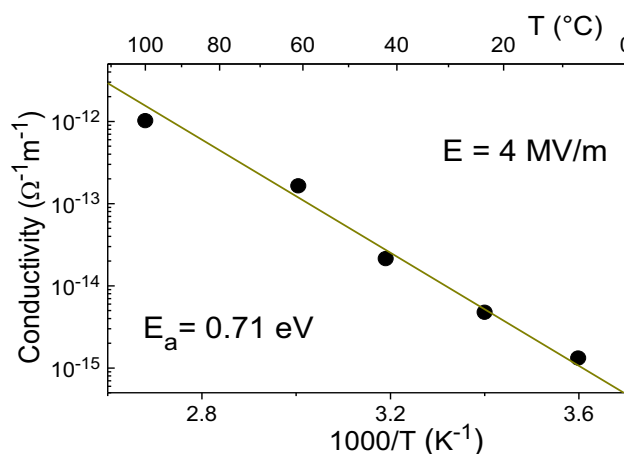


Figure 6. Arrhenius plot of conductivity vs. charging temperature for irradiated iPP at 100 kGy. Data are taken for a charging time of 3000 s.

$$J(t) = J_0(t) + A \sin(2\pi(t - t_0)/\Delta t) \quad (3)$$

where A is the amplitude, Δt is the period, $J_0(t)$ is the normally decaying current considered as a power law, and t_0 is the phase shift. Table 2 shows the values of the transit time, supposed equal to the period of the oscillations, as a function of the applied field. It is observed that the periods of the oscillations are roughly inversely proportional to the electric field. Also, the transit time appears to decrease with increasing the temperature. This behavior shows that these oscillations cannot be related to instabilities in the regulation system. The determined mobility value is around $\mu = 3 \times 10^{-14} m^2 V^{-1} s^{-1}$, which is of the same order of magnitude as that reported by other authors [33, 34] in the case of polyethylene. Oscillations in current have been correlated to repetitive charge packet formation notably by Alison [35]. The phenomenon of charge packet has been associated to the transport of holes through reorganization of the molecular microstructure under the influence of the electric field, featuring negative differential mobility in some instances [36, 37]. The point is that features reported in the present work are at much lower field, not likely to produce a direct effect on the molecular structure. So the processes at play here have probably not the same origin.

Ieda et al [38] have investigated the oxidation effect in the LDPE and they observed that the packet like space charge is accompanied by oscillatory conduction current. In addition, the increase in the temperature tends to lower the thresholds field which allows visualizing the charge packet. This appearance of the charge packet may be favored by the deterioration under the irradiation and by the fact that measurements are carried out at relatively high temperature. In this case, the carriers are easily supplied by thermal dissociation and injection, and they are easily accelerated [38]. The space charge packets crossing the insulation thickness from one electrode to the other have already been reported [39]. They are generally activated using very high DC fields or by relatively low fields in materials containing residues [37]. The phenomenon of the charge packets was observed for polyethylene [39] and for the cross-linked polyethylene [40].

3.4 THE FINAL THERMALLY STIMULATED DISCHARGING CURRENT

The FTSDC step was applied in order to analyze the trapped charges stored in shallow and / or deep traps during the ICC step and not released during the IDC.

The transferred charges can be localized in surface or bulk states by thermally and field activated processes. During the sample heating, the measured current can give information on the emptying of charged traps. Two parameters were exploited: the charging temperature T_C , and charging/discharging time ($t_c = t_d$).

3.4.1 EFFECT OF THE CHARGING TEMPERATURE

Figure 8 shows the final discharging currents of irradiated and virgin iPP in the temperature range between T_C and 120 °C, where T_C is varied from 2 to 60 °C. For these measurements, the heating rate ($b = 3$ °C/min), the electric field ($E = 4$ MV/m), and the charging and discharging times ($t_c = t_d = 3000$ s) were the same. Peaks in current are observed for the irradiated sample while in contrast, no peak was observed in the case of virgin iPP for the same conditions. The peak temperature increases as the charging temperature increases, see Table 3. In a general way, TSDC peaks are attributed to charge detrapping and transport or to dipoles reorientation. Banford et al [41] have also observed a very weak –insignificant– signal in TSDC in the virgin PP compared to the irradiated one at a dose of 38.6 kGy. The TSDC peak for the irradiated material was reported at about 330 K (37°C) for a charging temperature of 77°C. The response was interpreted rather as resulting from dipolar processes, given the features of impedance spectroscopy results. Alternatively, the temperature increase may release some trapped charges, which would be reflected by the appearance of current peak. Depending on the process, either oriented dipoles or trapped space charge would not be released from the sample during the isothermal short-circuiting and require the temperature increase. Along the orientation polarization scheme, irradiation would produce polar groups into the material as has been revealed experimentally [40]. Gamma radiations may also create charge traps into the iPP material.

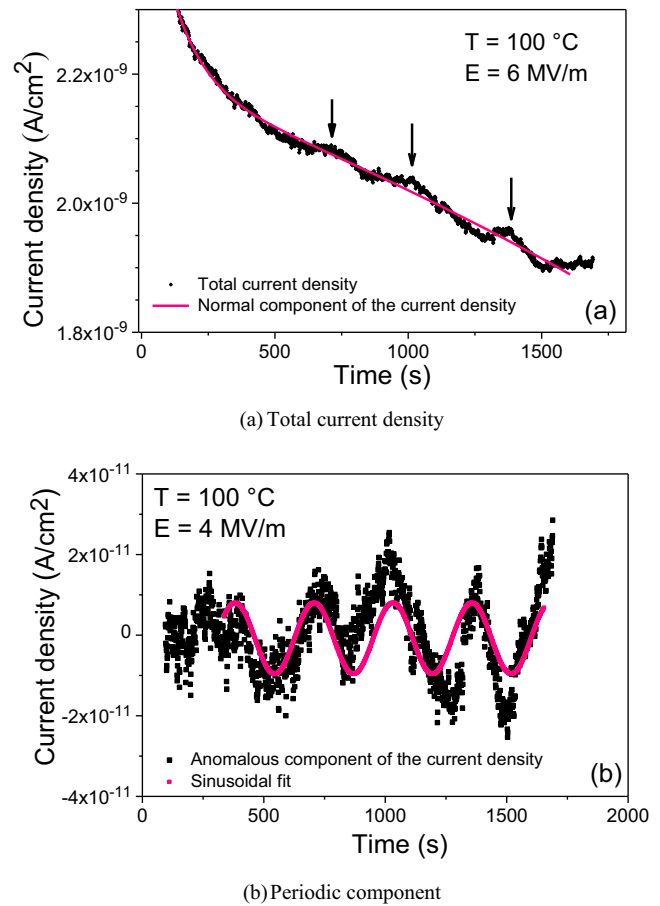


Figure 7. Time dependence of the charging current and of the periodic component of the current of the irradiated iPP at $T = 100$ °C.

Table 2. Parameters extracted from the current oscillation periods for $T = 100$ °C.

Electric field (MV/m)	2	4	6
Transit time (s)	500	400	310

During the irradiation process, the formation of some defects may occur in the amorphous or crystalline regions of the iPP films. The FTSDC curves contain a peak whose position changes with the charging temperature followed by a nearly exponential increase in current to the high temperature range. This feature may result from a distribution in relaxation time in case of dipolar response, or from a distribution in trap depth for the trapping scheme. The area under the TSDC peak provides information on the strength of the relaxation in case of dipolar response. It is related to the amount of charges involved in depolarization, and also to their nature and position in the insulation relative to the electrodes. Though it represents an underestimation of the total amount of detrapped charges [42], we used it as an estimate of charge quantity as:

$$Q = \int_{t_i}^{t_f} J(t) dt = \frac{1}{b} \int_{T_C}^{T_F} J(T) dT \quad (4)$$

where J is the current density and b is the heating rate. A Gaussian fitting of the FTSDC glow curve was achieved in

order to separate the peak from the exponential increase in current.

The parameters extracted from the curves after a Gaussian fit are listed in Table 3, where Q represents the charge related to the main peak. As T_C is increased to 60 °C, the apparent discharge peak becomes very large and shifts substantially to about 120 °C as if it corresponds to a different process. Similar results were observed in polyimide (PI) in the temperature range (20 – 350 °C) [43]. They were interpreted as due to the occurrence of a higher temperature relaxation process. The peak intensities as well as their characteristic temperatures increase with increasing charging temperature. Lança et al [44] also found the shift of the FTSDC curves at higher temperatures when the charging temperature was increased. More generally, the temperature of the apparent first peak is dependent on T_C , t_c , t_d and the heating rate [43].

3.4.2 EFFECT OF THE CHARGING/DISCHARGING TIME

The important increase of the current amplitude that appears after the peak at about 60 °C in the irradiated iPP (Figure 8), led us to explore the higher temperatures range. Thus, the depolarization current was recorded over a larger temperature range, from $T_C = 40$ °C to $T_F = 140$ °C. The charging /discharging time ($t_c = t_d$) is an important parameter that brings flexibility for the control of the activated dipoles according to their relaxation time and/or the energy and spatial distributions of the trapped charges. The FTSDC curves recorded at different charging times ($t_c = t_d = 3000$ s and 2000 s) are shown in Figure 9. The FTSDC spectra exhibit two peaks. The first one (P1) is centered at $T_1 = 75$ °C and the second one (P2) appears at higher temperature $T_2 \sim 122$ °C, which is close to the crystallization temperature. These peaks correspond to two relaxation processes from the polymer or from the release of charges from deep and shallow states. Moreover, the increase of the charging time t_c (from 2000 s to 3000 s) leads to an increase of peak P2 intensity, whereas the amplitude of the first one remains unchanged. The experimental data can be properly fitted by a superposition of two Gaussian profiles. The different parameters extracted from the FTSDC curves are listed in Table 4. By comparing the FTSDC spectra for charging times t_c of 2000 s and 3000 s, it can be noticed that the increase of the charging time did not have a substantial effect on peak P1 as its intensity and position only slightly change. The variation in the charges amount of peak P1 for different charging times t_c is only 4%. Contrarily, the intensity of peak P2 increases with increasing charging time and the ratio in peak area is about 3.5.

4 DISCUSSION

4.1 ORIGIN OF THE TSDC PEAKS

According to literature data, the relaxation spectrum of virgin iPP as a function of temperature is relatively well known; it exhibits four relaxation regions, which, for the sake of simplicity and practical reasons, are labeled as α , β , γ and δ , in order of decreasing temperature [20, 45-51]. Polyolefins, including PP, exhibit a dielectric relaxation (δ) at very low temperature (< -200 °C) attributed to hindered rotation of CH groups or to quantum mechanical tunneling processes related

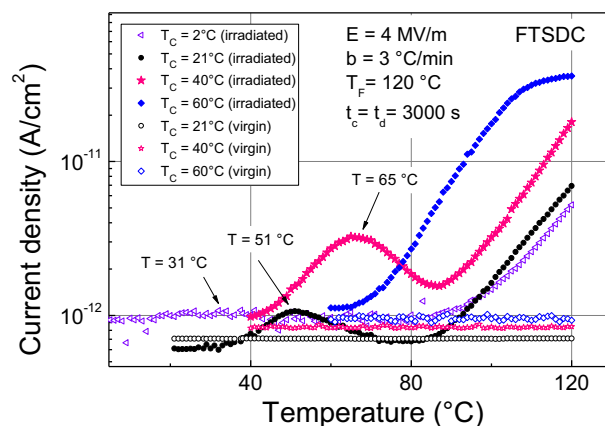


Figure 8. The final discharging current as a function of charging temperature for the irradiated and virgin iPP. The samples were previously charged for 3000 s followed by 3000 s of isothermal discharge at T_C . The final temperature was $T_F = 120$ °C.

Table 3. Parameters extracted from the FTSDC curves of Fig. 8. Quantities obtained with an electrode of 3.1 cm² area.

Charging Temperature (°C)	T_{MAX} (°C)	Charge Q (pC/cm ²)	Peak height (pA/cm ²)
2	31	1.6	0.13
21	51	10.6	0.45
40	65	47.4	2.16
60	120	/	32.3

to hydroxyl groups either on a stabilizer or on the polymer [46, 47]. The γ peak, according to different literature data and conditions, appears between -120 and -40 °C. This relaxation varies little with crystallinity and has been attributed to the local crank-shaft type motions in the amorphous phase [51]. The intermediate β peak situated between -25 and 25 °C has been identified with the glass transition of the amorphous fraction of iPP. All of those relaxation processes are found below room temperature at low frequency (1 Hz). The α process is localized below the melting temperature of PP. It is localized at 75 °C at 3 Hz by mechanical measurements [52]. The genesis of this process is controversial in the literature, but it undoubtedly requires the presence of a crystalline phase. According to Jourdan et al. [20], it is due to a relaxation of defects in the crystalline phase, and the amorphous phase in the neighborhood of the crystallites also contributes to the α process induced by migration of defects.

The above processes clearly correspond to relaxation phenomena associated to some chain motions / local reorganization in the material. The P1 peak observed herein is presumably related to the α -relaxation process. However, it does not mean that the response is dipolar in nature since molecular motions may favor the release of electrical charges.

The ionizing radiations to which samples were submitted provide the conditions for both polar groups formation and charge traps creation. Indeed, the radiation increases the amount of oxidized groups as carbonyl, hydroperoxide and other polar groups in non-polar iPP, which can act as dielectric probes of molecular motions due to the increase in polymer polarizability [53]. It may also be argued that polar groups

generate shallow traps and enhance charge injection [54, 55]. In many works, the TSDC response is interpreted in terms of charge release. For example, Ikezaki and Murata [56] clearly show a TSDC peak at 60 °C in ethylene propylene copolymer electrets that seems to correspond to the temperature range of the α -relaxation (or P1-peak). Materials were corona-charged in air, and because of the absence of any treatment that could make materials polar, the response is attributed to charge release.

A higher temperature peak at 120°C, which appears to correspond to P2 peak was reported by Ikezaki et al. [56]. The peak was described by a single set of traps with the same trap depth and the same capture cross-section. Iwazaki et al. [57] have attributed the appearance of the P2 peak (at higher temperature) to the defects in the crystalline region (forming deep traps).

The appeared peak at lower temperature (P1) in PP was also observed by TSDC technique in the range 60 to 80 °C [58 - 59]. A notable feature of the results presented therein is that the response appears non-linear in field [58, 60]. For this reason, we interpret the peaks in FTSDC as corresponding to the release of charges from traps, and the release can be favored by the molecular motions at play in the α -relaxation. P1 and P2 define two populations of traps, corresponding to shallow and deep traps, respectively. It is to note that whereas some relaxation processes can be well positioned, the FTSDC peak located near the charging temperature may correspond to partial activation of a distributed process, involving for example the shallower traps in a distribution.

4.2 RADIATION EFFECTS ON TRAP CHARACTERISTICS

A noticeable effect of irradiation of iPP is the reported disappearance of the low temperature (δ and γ) relaxations with gamma radiation. The complete “vanishing” of the γ relaxation in iPP irradiated in air is connected with a large radiation induced oxidative degradation in this material [61]. The question of oxidized groups effects on these local relaxation processes in irradiated polyolefins is not straightforward. On the one hand, it may produce a sensitization of the material due to the introduction of polar groups, making chain motion active dielectrically [62]. However, cross-linking and chain scission affect the mobility of macromolecules, especially in the amorphous phase, causing a shift of the relaxation maxima and a change in activation energy of dielectric relaxation to which the mentioned dipolar and molecular movements contribute [63]. In some instances as with iPP it may hinder the relaxation. While the γ -relaxation disappears in iPP due to oxidation or possibly to crosslinking, in polyethylenes being low or high density, the effect is opposite [63], albeit in presence of substantial oxidation. In case of the β -relaxation, the trend is an increase of the response with increasing irradiation of PP as observed by impedance spectroscopy [62] or TSDC [41] and again the effect is ascribed to oxidized groups formed on the main chains in the amorphous phase.

The above relaxation phenomena, reported for doses of some hundreds of kGy, presumably do not impact directly the

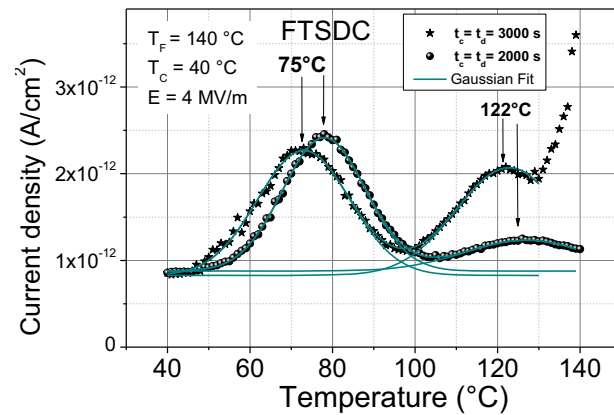


Figure 9. The FTSDC for different charging and discharging times in irradiated iPP and fit to Gaussian curves. The samples were previously charged then isothermally discharged at 40 °C, 4 MV/m with times as indicated in the caption.

Table 4. Parameters extracted from the FTSDC curves of Fig. 9

Peak number	Tmax (°C)	Charge Q (pC/cm²)	Width (°C)	Height (pA/cm²)
P1 ($t_c = 2000$ s)	77	39.0	20.1	1.55
P2 ($t_c = 2000$ s)	126	13.2	29.7	0.35
P1 ($t_c = 3000$ s)	72	40.6	22.3	1.45
P2 ($t_c = 3000$ s)	123	41.9	26.6	1.26

polarization behavior at room temperature and beyond at the time scales investigated in this work. However they let us anticipate that chemical modifications are introduced that are likely to impact transport processes. The evident effects observed in the present work are an increase in the conductivity of the material, and the appearance of oscillations in the current. These effects are presumably due to deep traps creation.

According to Yamashita et al. [58], the deep traps can exist in the crystalline region of PP, where acetone molecules cannot penetrate. They have concluded that, the high density of deep traps in the crystal surface region can be attributed to the high probability of crystalline defects production. In our case, the first factor which allows creating defects is related to the gamma radiations (the degree of crystallinity increases with the dose). Thus, if the application of electric field lasts longer, the charges occupy the deep traps created in the crystalline region. The crystalline defects are resulting from the impact of the gamma irradiations on iPP material.

With some hypotheses, the initial rise method, first advocated by Garlick and Gibson provides a good estimate of the depth of traps involved in the discharge current [64]. The method is based on the fact that the rise of the low-temperature side of a TSDC peak follows an Arrhenius law for temperatures well below the peak temperature. The apparent trap depth E_a was estimated from the rise of peaks P1 and P2 using this approach. The obtained values of E_a increase from 0.09 to 0.4 eV, when t_c increases from 2000 to 3000 s, cf. Figure 10. Similar variation of E_a vs. t_c was observed by Neagu et al [65]. It was found that the charging time influences the quantity of charges trapped during the

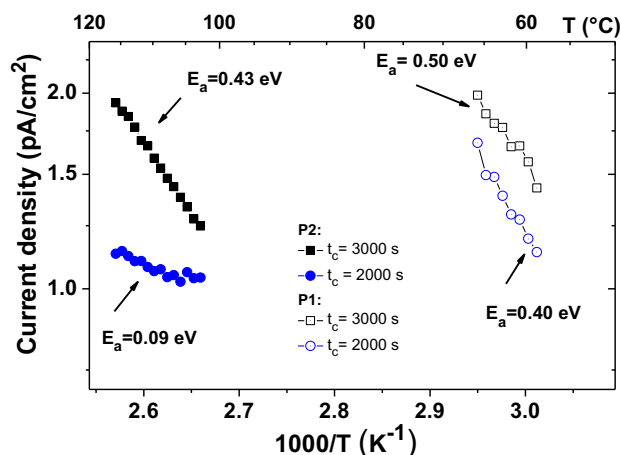


Figure 10. Activation energy deduced from the initial rise method for FTSDC peaks P1 and P2.

isothermal step (ICC) as well as the trap depth. It appears that the probability of trapping and the concentration of trapped charges in deep traps increase as the charging time (under field) increases. As during the isothermal discharging process, the trapped charges have not a sufficient energy to be released, more are detected during the last step (FTSDC). The fact that the amount of released charges for t_c and $t_d = 3000$ s is much larger than for 2000 s may find its origin in the longer time given for charges to trap, or to the trapping of charges deeper into the material. According to the analysis of the P1 peak, the activation energy is less dependent on the charging time owing to more efficient trap filling in the corresponding charging conditions. In both cases, the estimated depth of traps is relatively small. The formation of such relatively shallow traps is possibly related to the increase in conductivity observed after irradiation.

Further argument can be provided regarding the incomplete charging of the material in the conditions were results of Figure 9 were obtained: considering a transit time for charges of 400 s at 100 °C as deduced from the oscillatory currents, – cf. Table 2, and hypothesizing an activation energy for transport of 0.71 eV as found above for conduction, then the transit time at $T_c = 40$ °C would be about 400 min. This is substantially longer than charging times used here. Therefore, charge trapping would occur only in a part of the material thickness. Measurement of the position of charges would be worth addressing, but the small thickness of samples makes space charge measurements difficult to implement for such samples.

5. CONCLUSION

The aim of this work was to investigate the trapping/detrapping of the space charge in virgin and irradiated iPP by the non-isothermal discharging current technique coupled to physicochemical characterization. Fourier transform infrared spectroscopy (FTIR) analyses indicate an increase of the carbonyl index which implies an oxidation of the irradiated iPP. In this case, the radiation-induced oxidation gives rise to polar groups formation that are responsible for polymer chains motion. The presence of peaks

in the FTSDC spectrum of the irradiated sample is attributed to the detrapping of space charges from deep or shallow traps created upon irradiation. The increase of the charging time revealed two characteristic peaks, at low and high temperature. Moreover, the amount of trapped charges increases with the charging temperature T_c . A considerable increase of the degree crystallinity was measured by DSC after irradiation and this feature can be attributed to the cross-linking phenomenon. Consistently with previous reports, it was concluded that traps were mostly created in the defective crystalline regions of the material. In the next works, we will exploit the space charge distribution measurements to get a better understanding of the charges behavior in the bulk material.

ACKNOWLEDGMENT

The authors are thankful to the Director and the staff of the Nuclear Research Center of Algiers for the irradiation of the samples.

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